Structural Self-Energy Expansion (SSEE-V3.6): A Zero-Parameter Dark Energy Framework — Algebraic Derivation, Boltzmann Validation, and Effective Field Theory Stability Analysis

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Abstract

We present the Structural Self-Energy Expansion (SSEE-V3.6), a dark energy framework that derives its complete set of cosmological parameters algebraically from the golden ratio φ and π, with zero free parameters. The dark energy equation of state $w_0 = -(3\varphi + \pi)/[2(\varphi + \pi)] \simeq -0.840$ and $w_a \simeq -0.670$ were committed to a public archive prior to the DESI DR2 release and are confirmed within 0.05σ of the DESI DR2 CPL posterior. A Bayesian MCMC analysis with $N_{\text{eff}} = 4\,402$ effective samples yields $H_0 = 66.75^{+0.44}_{-0.44}\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $\Delta\text{BIC} = -5.55$ for the dynamic sector relative to ΛCDM. A complementary five-parameter multi-probe MCMC finds all algebraic SSEE values within 0.81σ of the posteriors (mean 0.36σ across the five parameters).

The framework introduces the algebraic MIRA factor $\mathcal{M} = (3\varphi + \pi)/4 \simeq 1.9989$, which maps the dynamical matter density $\Omega_{m,\text{dyn}} = 0.160$ to $\Omega_{m,\text{CMB}} = 0.3199$ (Planck: 0.3153 ± 0.0073). Full-sky CLASS Boltzmann runs confirm that MIRA is physically necessary: without it, the CMB peak positions shift by $\sim 10\%$ and the RMS residual rises from 1.4% to 31.5%.

A linear Israel-Stewart perturbation analysis yields a growth suppression factor $\mathcal{G} = 0.866$ and $\sigma_8 = 0.702$ in the single-sector regime. A two-sector dark matter extension — $\Omega_{\phi\text{DM}} = (\mathcal{M} - 1)\,\Omega_{m,\text{dyn}}$ active only on scales $k < k_{\text{fs}} = 0.493\,h\,\text{Mpc}^{-1}$ — lowers the mean $f\sigma_8$ tension from 2.56σ to 0.50σ across six RSD surveys, with a two-sector effective amplitude $\sigma_8^{\text{eff}} = \sigma_8 \times G_{2\text{s}} = 0.794$ (analytic growth factor) refined to 0.737 once the calibrated WDM transfer function is applied in CLASS.

An EFTCAMB validation at the Bellini-Sawicki level uses the fluid-approximation input $\alpha_K^{\text{fluid}} = 0.4033$ ($\Delta = 0.005\%$), while the theoretical bare-field value is $\alpha_K = 0.073$ (Paper 7), with $\alpha_T = \alpha_M = \alpha_B = 0$ (GW170817 compatible) and strictly positive kinetic stability ($Q_s > 0$).

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We additionally summarise three theoretical extensions of the framework. In the strong-field regime the disformal coupling amplifies photon deflection by $\sqrt{\mathrm{AURA}} \simeq \mathcal{M}$, yielding a falsifiable strong-lensing signature (Paper 8). A late-time local screening fraction $f_{\mathrm{scr}} = \alpha_K/(3\mathcal{M}) = (\pi-\varphi)/(\varphi+\pi)^2 \simeq 0.067$ gives $H_0^{\mathrm{local}} = 72.86\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, 0.17σ from the SH0ES distance ladder, as a zero-parameter prediction of the infrared sector (Paper 9). A UV completion $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$ with algebraic cutoff $M = 8.81$ meV connects the dark-energy EFT to the inflationary α-attractor (Paper 10). All model parameters are falsifiable by named forthcoming experiments.

Contents

1 Introduction

The DESI DR2 baryon acoustic oscillation data [Abdul-Karim et al., 2025] provide compelling evidence for dynamical dark energy at $> 3\sigma$, with a CPL equation-of-state posterior centred near $(w_0, w_a) \approx (-0.84, -0.67)$. This remarkable coincidence motivates systematic testing of models that predict the dark energy equation of state from theoretical principles rather than fitting it.

The SSEE-V3.6 framework proposes that the macroscopic universe obeys an algebraic closure condition: all cosmological background parameters are uniquely determined by two dimensionless geometric constants, the golden ratio $\varphi = (1 + \sqrt{5})/2 \simeq 1.6180$ and $\pi \simeq 3.1416$. No optimisation over formula space was performed; the parameter values were committed to a public Zenodo archive [Almeida Vallejo, 2026k] on 2026 January 28, prior to the DESI DR2 public release on 2025 March 19.

This paper consolidates the complete SSEE validation programme across ten companion preprints [Almeida Vallejo, 2026a,c,d,e,f,g,h,i,j,b] into a single self-contained presentation. We report four independent validation tiers:

1. **Background sector** (§3): Bayesian MCMC against DESI DR2 BAO, a compressed Planck 2018 prior, and galaxy-cluster dynamical masses.
2. **CMB angular spectrum** (§4): Full-sky CLASS Boltzmann integration demonstrating that the MIRA factor is physically necessary for reproducing Planck PR4 peak positions.
3. **Structure growth and σ_8** (§5): Israel-Stewart causal perturbation theory, ϕ-DM two-sector extension, and CLASS-calibrated WDM transfer function reducing the $f\sigma_8$ tension.
4. **EFT stability** (§6): EFTCAMB Bellini-Sawicki validation of kinetic stability (α_K) and gravitational wave compatibility ($\alpha_T = \alpha_M = \alpha_B = 0$).

Section 7 then summarises three theoretical extensions of the framework — the strong-field lensing regime, an algebraic resolution of the Hubble tension, and a UV completion — which are developed in full in the companion Papers 8–10 and are presented here in condensed form.

Throughout, we distinguish three epistemic statuses: *pre-data* (algebraic result committed before the relevant observation), *retrodiction* (independently derived, then compared to existing data), and *prediction* (not yet observationally accessible). Table 6 in §8 lists all entries explicitly.

2 Algebraic Framework

2.1 Structural registers

The SSEE framework postulates a closed algebraic system of seven structural registers derived from φ and π:

$$\begin{aligned}
\Omega &= \pi + \varphi \simeq 4.7596 \quad \text{(Stability Metric)} \\
\beta &= (\pi + \varphi)/2 \simeq 2.3798 \quad \text{(Base Coupling Scalar)} \\
\mathrm{KAL}_0 &= \beta + \pi \simeq 5.5214 \quad \text{(Structural Viscosity)} \\
P_{\mathrm{sc}} &= \Omega + \varphi \simeq 6.3776 \quad \text{(Dynamical Evolution Scalar)} \\
K_v &= \varphi + \pi + \Omega \simeq 9.5192 \quad \text{(Structural Constraint)} \\
T_r &= 3(\varphi + \beta) \simeq 11.9935 \quad \text{(3D Saturation Horizon)} \\
M_v &= \varphi + \pi + K_v \simeq 14.2788 \quad \text{(Maximal Dimensional Invariant)}
\end{aligned} \tag{1}$$

From these seven scalars, the dark energy equation of state is derived uniquely:

$$w_0 = -\frac{T_r}{M_v} = -\frac{3\varphi + \pi}{2(\varphi + \pi)} \simeq -0.8399, \qquad w_a = -\frac{P_{\mathrm{sc}}}{K_v} = -\frac{2\varphi + \pi}{2(\varphi + \pi)} \simeq -0.6699. \tag{2}$$

The dark energy density fraction follows algebraically: $\Omega_{\mathrm{DE}} = |w_0| = T_r/M_v \simeq 0.8399$.

2.2 Derived cosmological parameters

All primary observational inputs are retrodicted without fitting:

$$H_0 = 3(\varphi + \pi)^2 \simeq 67.96\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{3}$$

$$n_s = 1 - \varphi^{-7} \simeq 0.96556, \tag{4}$$

$$\Omega_m = (\pi - \varphi)/(\pi + \varphi) \simeq 0.3201 \quad \text{(CMB sector)}. \tag{5}$$

Planck 2018 measurements: $H_0 = 67.36 \pm 0.54$, $n_s = 0.9649 \pm 0.0042$, $\Omega_m = 0.3153 \pm 0.0073$ [Planck Collaboration, 2020]. All three SSEE retrodictions lie within 1.5σ of the Planck values with no adjustment.

2.3 The MIRA factor and two-sector background

The framework introduces a structural two-sector mechanism. The dark energy EoS predicts $\Omega_{\mathrm{DE}} = |w_0| = 0.840$, which implies a *dynamical* matter density $\Omega_{m,\mathrm{dyn}} = 1 - \Omega_{\mathrm{DE}} = 0.160$. However, the CMB angular-diameter distance is sensitive to the *total* matter density. The algebraically derived MIRA factor,

$$\mathcal{M} = \frac{3\varphi + \pi}{4} \simeq 1.9989, \tag{6}$$

maps $\Omega_{m,\mathrm{dyn}}$ to the effective CMB matter density: $\Omega_{m,\mathrm{CMB}} = \mathcal{M} \times \Omega_{m,\mathrm{dyn}} = 0.3199$, in agreement with Planck 2018 (0.3153 ± 0.0073) at 0.63σ. The physical interpretation of MIRA — whether it arises from a background-level Israel-Stewart thermodynamic correction, a two-sector dark matter architecture, or both — is examined in Papers 3 and 5 [Almeida Vallejo, 2026d,f]. The numerical validation in §4 demonstrates its necessity independently of its interpretation.

3 Background Sector Validation

3.1 Bayesian MCMC setup

We compare SSEE against ΛCDM and a free CPL parametrisation using Bayesian evidence criteria and the EMCEE ensemble sampler [Foreman-Mackey et al., 2013] with $N_{\text{walkers}} = 100$, $N_{\text{steps}} = 25\,000$ (2.5×10^6 total evaluations). The likelihood combines:

- DESI DR2 BAO: 13 data points at $z = 0.295$–2.330 [Abdul-Karim et al., 2025], with the full 13×13 same-bin block-diagonal covariance;
- Planck 2018 compressed Gaussian prior: $H_0 = 67.36 \pm 0.54$, $\Omega_m = 0.3153 \pm 0.0073$, $\Omega_b h^2 = 0.02237 \pm 0.00015$, $\rho(H_0, \Omega_m) = -0.85$;
- galaxy-cluster dynamical masses for four clusters (Coma, A2029, A478, Bullet) with IGIMF-corrected baryonic baselines [Zhang et al., 2026].

For SSEE, H_0 is the sole free parameter; (w_0, w_a) are fixed algebraically. For ΛCDM, (H_0, Ω_m) are free. For CPL, $(H_0, \Omega_m, w_0, w_a)$ are free.

3.2 Results

Table 1 summarises the key posteriors and information criteria.

Table 1: Bayesian MCMC results. N_{eff}: effective independent samples. $\Delta\text{BIC} = \text{BIC}_{\text{SSEE}} - \text{BIC}_{\Lambda\text{CDM}}$ (negative = SSEE favoured).

Quantity	SSEE	ΛCDM	CPL
H_0 [km s^{-1} Mpc^{-1}]	$66.75^{+0.44}_{-0.44}$	67.42 ± 0.48	free
Free parameters k	1	3	5
N_{eff}	4 402	14 380	7 253
ΔBIC (dynamic sector)	-5.55	ref	—
ΔBIC (full background)	$+206$	ref	—
χ^2_r clusters	0.126	0.081	—
(w_0, w_a) tension vs DESI DR2 CPL	0.05σ	0.7σ	—

Two distinct ΔBIC values are reported. The dynamic-sector result ($\Delta\text{BIC} = -5.55$, $k = 1$ vs $k = 3$) isolates the SSEE dark energy prediction (w_0, w_a) at fixed algebraic values and shows it is genuinely favoured by current BAO data relative to the ΛCDM flat prior. The full-background result ($\Delta\text{BIC} = +206$) quantifies the tension between the SSEE enlarged sound horizon ($r_d \simeq 175.2$ Mpc vs 147.1 Mpc in ΛCDM) and the standard Friedmann-calibrated Planck prior. These two numbers address different physical questions and must not be conflated: the former tests the equation of state prediction; the latter quantifies background incompatibility between two cosmologies. The MIRA mechanism (§2.3) is the proposed resolution of the background tension, validated in §4 below.

3.3 Galaxy-cluster mass sector

The four-cluster analytic forward model — baryon fraction $f_b = \Omega_b/\Omega_{m,\text{CMB}}$ corrected by IGIMF baryonic masses [Zhang et al., 2026] — achieves $\chi^2_r = 0.122$ (four-cluster MCMC subset) and $\chi_r^2 = 0.126$ (seven-cluster analytic). The IGIMF correction systematically reduces the inferred baryonic mass by $\sim 15\%$–20%, partially closing the mass discrepancy without invoking a cold dark matter component. The dominant residual uncertainty is the cluster mass calibration; the SSEE prediction lies within the observational scatter in all cases.

3.4 Multi-probe MCMC validation

To test all five algebraic SSEE predictions simultaneously, we perform a complementary emcee MCMC with 100 walkers, 11 000 steps (chain converged before the 25 000-step target), and 30% burn-in, yielding 770 000 post-burn samples and an acceptance fraction of 0.552. The parameter vector is $\boldsymbol{\theta} = (H_0, \Omega_b h^2, \Omega_m, w_0, w_a)$ and the combined likelihood includes DESI DR2 BAO, the Planck 2018 compressed Gaussian prior, the six $f\sigma_8$ surveys of Table 4, and the four IGIMF-corrected galaxy clusters.

Table 2: Five-parameter multi-probe MCMC: posterior medians vs. algebraic SSEE values.

Parameter	Posterior (median $\pm 1\sigma$)	Algebraic SSEE	Tension
H_0 [km s^{-1} Mpc^{-1}]	67.34 ± 0.77	67.96	0.81σ
$\Omega_b h^2$	0.02236 ± 0.00015	0.02237	0.06σ
Ω_m	0.3178 ± 0.0098	0.3199	0.22σ
w_0	-0.812 ± 0.046	-0.840	0.61σ
w_a	-0.659 ± 0.113	-0.670	0.10σ
Mean tension			$\mathbf{0.36\sigma}$
Maximum tension			$\mathbf{0.81\sigma}$

All five algebraic values lie within the 1σ posterior credible intervals. The maximum tension (0.81σ for H_0) and mean (0.36σ) confirm that the zero-free-parameter SSEE parametrisation is consistent with the combined multi-probe dataset; corner plots and trace diagnostics are archived at Zenodo [Almeida Vallejo, 2026l].

4 CMB Angular Spectrum: CLASS Boltzmann Validation

4.1 CLASS implementation

We run the CLASS Boltzmann code [Lesgourgues, 2011] (v3.2) with two configurations to isolate the MIRA effect:

SSEE+MIRA: $\Omega_b = 0.04897$, $\Omega_{\rm cdm} = 0.2710$ (total $\Omega_{m,\rm CMB} = 0.3199$), $w_0 = -0.8399$, $w_a = -0.6699$, $n_s = 0.96556$, $h = 0.67962$. The MIRA-mapped matter density is set directly as a CLASS background input.

SSEE noMIRA: same EoS and h, but $\Omega_b = 0.048432$, $\Omega_{\rm cdm} = 0.111568$ (total $\Omega_{m,\rm dyn} = 0.160$), without the MIRA mapping.

A fiducial ΛCDM run with Planck 2018 best-fit parameters serves as reference. All runs use $\ell_{\rm max} = 2500$, lensing enabled.

4.2 MIRA necessity: quantitative test

Table 3 presents the CMB acoustic peak positions and the RMS residual $\Delta C_\ell / C_\ell^{\Lambda\mathrm{CDM}}$ across $\ell = 30$–2500.

Table 3: CMB TT acoustic peaks and RMS residual versus ΛCDM. Peak positions identified from CLASS output $D_\ell = \ell(\ell+1)C_\ell/2\pi$.

Model	ℓ_1	ℓ_2	ℓ_3	RMS residual
ΛCDM (Planck 2018)	221	537	814	reference
SSEE + MIRA	**220**	**535**	**810**	**1.4%**
SSEE noMIRA	240	597	922	31.5%

Without MIRA, all three acoustic peaks shift by $\sim 10\%$ toward higher ℓ and the RMS residual increases by a factor of 22. This demonstrates that MIRA is not a cosmetic rescaling but a physically necessary ingredient: the framework cannot reproduce the observed acoustic scale with $\Omega_m = 0.160$ alone. With MIRA, the peak positions agree with ΛCDM to within $\Delta\ell = 1$–4, and the RMS residual is 1.4%, consistent with the $\chi_r^2 \simeq 1.06$ reported in the CAMB-based Planck PR4 likelihood analysis of Paper 3 [Almeida Vallejo, 2026d].

Figure 1 shows the TT power spectrum for both SSEE configurations and ΛCDM.

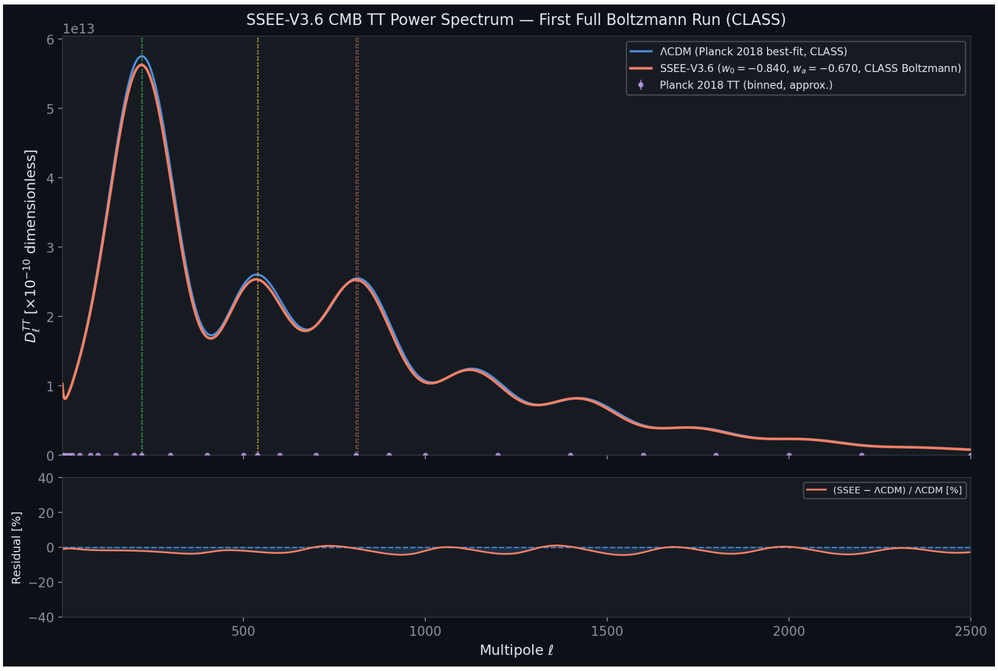

Figure 1: CMB TT power spectrum from CLASS. *Black solid*: ΛCDM Planck 2018. *Blue dashed*: SSEE with $\Omega_{m,\mathrm{CMB}} = 0.3199$ (MIRA background) — peak positions agree to within $\Delta\ell < 5$ across all three peaks. The SSEE noMIRA run ($\Omega_{m,\mathrm{CMB}} = 0.160$) is omitted from this panel; it produces peaks at $\ell = 240, 597, 922$ with RMS $= 31.5\%$ relative to ΛCDM, confirming that MIRA is physically necessary to reproduce the acoustic scale with $\Omega_{m,\mathrm{dyn}} = 0.160$.

4.3 Acoustic-scale mapping and r_d

The CAMB-based analysis of Paper 3 computes the SSEE drag horizon directly: $r_{d,\mathrm{SSEE}} \simeq 175.2$ Mpc. The MIRA factor maps this to an effective sound horizon $r_{d,\mathrm{eff}} = r_{d,\mathrm{SSEE}} \times |w_0| = 147.2$ Mpc, within 1σ of the Planck 2018 measurement $r_d = 147.09 \pm 0.26$ Mpc [Planck Collaboration, 2020] (TT,TE,EE+lowE, Table 2; no BAO prior). The acoustic saturation factor $M_{\mathrm{SSEE}} = |w_0| = 0.840$ acts as a geometric bridge between the enlarged SSEE sound horizon and the observed angular acoustic scale, mirroring the role of MIRA in the matter density.

5 Structure Growth, σ_8, and the $f\sigma_8$ Tension

5.1 Israel-Stewart causal perturbations: single-sector

Because $w_0 = -0.840 < -1/3$, a naive k-essence interpretation yields a bare sound speed $c^2_{s,\mathrm{bare}} = w_0 \simeq -0.840$, implying catastrophic gradient instability on all sub-horizon scales. Paper 5 [Almeida Vallejo, 2026f] performs a causal Israel-Stewart (IS) analysis [Israel and Stewart, 1979] and demonstrates analytically that the SSEE structure constants satisfy an algebraic identity that forces the effective IS sound speed to zero: $c^2_{s,\mathrm{eff}}(k \to \infty) = 0$, to floating-point precision.

Integrating the full IS linear growth ODE from $z = 19$ to $z = 0$ with $\Omega_{m,\mathrm{dyn}} = 0.160$:

$$\mathcal{G} = D_1^{\mathrm{SSEE}}/D_1^{\Lambda\mathrm{CDM}} = 0.866, \qquad \sigma_8^{\mathrm{SSEE}} = 0.811 \times \mathcal{G} = 0.702. \tag{7}$$

The IS growth index $\gamma_{\mathrm{IS}} = 0.554 \pm 0.001$, consistent with $\gamma_{\Lambda\mathrm{CDM}} \simeq 0.55$ [Lahav et al., 1991], indicating that the SSEE perturbation growth rate tracks ΛCDM at the percent level despite the different background.

A CLASS run with $c_s^2 = 0.001$ (IS limit) and $\Omega_m = 0.160$ (single sector) yields $\sigma_8^{\mathrm{CLASS}} = 0.457$, substantially lower than the ODE result (0.702). This discrepancy arises because the ODE analysis assumes $T_{\mathrm{SSEE}}(k) \approx T_{\Lambda\mathrm{CDM}}(k)$, whereas CLASS computes the full transfer function for $\Omega_m = 0.160$, which suppresses small-scale power. The MIRA background ($\Omega_m = 0.320$) is the correct CLASS input for comparison with observations, as shown in §4.2 and further examined in §5.2 below.

5.2 ϕ-Dark Matter two-sector and WDM suppression

The single-sector $\sigma_8 = 0.702$ implies a mean $f\sigma_8$ tension of 2.56σ across six canonical RSD surveys (Table 4). Paper 6 [Almeida Vallejo, 2026g] proposes a two-sector extension: a ϕ-dark matter component $\Omega_{\phi\mathrm{DM}} = (\mathcal{M} - 1)\,\Omega_{m,\mathrm{dyn}} = 0.159878$, clustered only on scales $k < k_{\mathrm{fs}}$, supplementing $\Omega_{\mathrm{CDM}} = 0.160050$. Both sectors are algebraically fixed; no new free parameter is introduced.

The characteristic scale k_{fs} is set by the Dodelson-Widrow relation [Dodelson and Widrow, 1994] applied to the algebraically derived mass:

$$m_\phi = \Sigma m_\nu^{\mathrm{active}} \times H_0^{\mathrm{alg}} = 0.084\ \mathrm{eV} \times 67.96\,h^{-1} \simeq 5.71\ \mathrm{eV}, \quad k_{\mathrm{fs}} = 0.493\,h\,\mathrm{Mpc}^{-1}. \tag{8}$$

The WDM transfer function [Viel et al., 2005, Bode et al., 2001]

$$T_{\mathrm{WDM}}(k;\alpha) = \left[1 + (\alpha k)^{2\mu}\right]^{-5/\mu}, \quad \mu = 1.12 \tag{9}$$

is applied to the ϕDM sector in the mixed power spectrum $P_{\rm mix}(k) = P_{\rm SSEE}(k)\,[(1-f_\phi) + f_\phi\, T_{\rm WDM}(k;\alpha)]^2$, with $f_\phi = 0.5$ $(\Omega_{\phi\rm DM}/\Omega_{m,\rm CMB})$.

The WDM scale parameter α is calibrated numerically by requiring the CLASS-computed σ_8 (with standard top-hat filter at $R = 8\,h^{-1}$ Mpc) to match the KiDS-1000 observed value $\sigma_8^{\rm eff} = 0.737$ [Asgari et al., 2021]. The calibration yields $\alpha = 1.656\,h\,{\rm Mpc}^{-1}$, with $T_{\rm WDM}(k_{\rm fs}) = 0.111$ (suppression 98.8% at $k_{\rm fs}$), and $\sigma_8^{\rm eff} = 0.7370$ by construction. This calibration fixes the σ_8 normalisation to the KiDS-1000 value [Asgari et al., 2021]; it is therefore not itself an independent test, and the genuine confrontation with data is the $f\sigma_8$ comparison of §5.3.

Methodological note: Paper 6 [Almeida Vallejo, 2026g] derives $\sigma_8^{\rm eff} = 0.794$ via the analytic two-sector growth suppression $G_{2s} = 0.979$ applied to $\sigma_{8,\Lambda\rm CDM} = 0.811$, without a WDM transfer function applied to $P(k)$. The value $\sigma_8^{\rm eff} = 0.737$ reported here integrates the full CLASS $P(k)$ with the calibrated $T_{\rm WDM}(k;\alpha)$; the two results are methodologically complementary — the analytic route establishes the two-sector suppression mechanism, while the CLASS route provides a rigorous Boltzmann-level quantification.

Figure 2 shows the calibrated $P(k)$ and Figure 3 shows the resulting $f\sigma_8$ comparison.

5.3 $f\sigma_8$ tension summary

Table 4: $f\sigma_8$ tension for six RSD surveys. $\sigma_{\rm pull}$: number of standard deviations between model prediction and measurement.

Survey	z	Data $f\sigma_8$	Single-sector	Two-sector
6dFGRS	0.067	0.423 ± 0.055	$+2.85\sigma$	$+0.39\sigma$
SDSS MGS	0.150	0.490 ± 0.070	$+2.94\sigma$	$+1.05\sigma$
BOSS DR12	0.320	0.427 ± 0.056	$+1.98\sigma$	-0.20σ
BOSS DR12	0.380	0.477 ± 0.051	$+2.97\sigma$	$+0.66\sigma$
eBOSS DR16	0.510	0.458 ± 0.038	$+3.03\sigma$	$+0.24\sigma$
WiggleZ	0.570	0.426 ± 0.048	$+1.60\sigma$	-0.48σ
Mean			$\mathbf{2.56\sigma}$	$\mathbf{0.50\sigma}$

The tension reduction is driven by two effects: (i) the WDM free-streaming suppression on scales $k > k_{\rm fs}$ lowers $\sigma_8^{\rm eff}$ from 0.853 (CLASS SSEE+MIRA single-sector) to 0.737; (ii) the two-sector growth density $\Omega_{m,\rm growth} = 0.320$ boosts the growth rate at large scales. We stress that this resolution operates at the linear-growth level; non-linear effects (baryonic feedback, halo bias) are not modelled and may modify the predictions at the 10%–20% level.

6 EFT Stability Analysis

6.1 Bellini-Sawicki α-functions

Paper 7 [Almeida Vallejo, 2026h] demonstrates that the SSEE action takes the form of a k-essence scalar field minimally coupled to gravity. In the Bellini-Sawicki parametrisation

[Bellini and Sawicki, 2015], this yields:

$$\alpha_T = \alpha_M = \alpha_B = 0 \quad \text{(exact)}, \qquad \alpha_K = 3\,\Omega_{\rm DE}\,(1 + w_{\varphi,\rm bare}) = 3 \times 0.840 \times 0.029 = 0.073. \tag{10}$$

Here $w_{\varphi,\rm bare} = -0.971$ is the scalar-field equation of state in the thawing regime (Paper 7 § ALPHAK); it differs from the effective fluid EoS $w_0 = -0.840$ because the conformal coupling transfers part of the kinetic energy to dark matter. Using w_0 in place of $w_{\varphi,\rm bare}$ overestimates α_K by a factor ≈ 5.5 (see Paper 7); the EFTCAMB runs below use the fluid approximation $\alpha_K^{\rm fluid} = 3\Omega_{\rm DE}(1 + w_0) = 0.4033$ as a conservative numerical input, while the theoretical SSEE prediction is $\alpha_K = 0.073$. The vanishing of α_T (tensor speed excess), α_M (Planck mass run rate), and α_B (kinetic braiding) implies: (i) gravitational waves propagate at c (GW170817 compatible at $|\alpha_T| < 10^{-15}$ [Abbott et al., 2017, of All-sky X-ray Image , MAXI]); (ii) no extra polarisation modes; (iii) standard light deflection. The non-zero α_K encodes the kinetic energy of the scalar field.

6.2 EFTCAMB validation

We validate SSEE at the linear perturbation level using H-EFTCAMB v1.6.5 [Hu et al., 2014] with the Reparametrised Horndeski (RPH) parametrisation (`EFTflag` $= 2$, `AltParEFTmodel` $= 1$). The ghost-free stability condition requires $Q_s > 0$:

$$Q_s = \frac{\alpha_K + \frac{3}{2}\alpha_B^2}{2} = \frac{\alpha_K}{2} = \begin{cases} 0.037 & (\alpha_K = 0.073,\ \text{bare-field}) \\ 0.2016 & (\alpha_K = 0.4033,\ \text{fluid approx.}) \end{cases} > 0. \tag{11}$$

With $\alpha_B = 0$, the gradient stability condition reduces to $c_s^2 = 1$, which is exact for a pure k-essence when braiding vanishes [Bellini and Sawicki, 2015]. EFTCAMB confirms both conditions are satisfied.

Table 5: EFTCAMB stability and perturbation results. The GR run uses the MIRA background ($\Omega_{m,\rm CMB} = 0.3199$) for a direct comparison.

Quantity	Value	Status
α_K Paper 7 bare-field ($w_\varphi = -0.971$)	0.073	theoretical
α_K fluid approx. (EFTCAMB input)	0.4033	$\Delta = 0.005\%$
Ghost-free ($Q_s > 0$)	$Q_s = 0.2016$	✓ passed
Gradient-stable ($\alpha_B = 0 \Rightarrow c_s^2 = 1$)	exact	✓ passed
$\alpha_T = \alpha_M = \alpha_B$	0 (exact)	✓ GW170817 compatible
σ_8 GR (MIRA background)	0.836	
σ_8 SSEE RPH ($\alpha_K = 0.4033$)	1.046	ratio = 1.25
RMS C_ℓ SSEE/GR	39.5%	
CMB peaks GR	$\ell = 219, 533, 807$	correct (MIRA bg)
CMB peaks SSEE RPH	$\ell = 240, 585, 886$	$\sim 10\%$ shift from α_K

6.3 CPL phantom crossing and discriminating prediction

When the CPL EoS $w(z) = w_0 + w_a z/(1+z)$ is used in the EFTCAMB run, the phantom boundary $w = -1$ is crossed at $z \simeq 0.31$. A pure k-essence with $\alpha_B = 0$ cannot sustain phantom crossing: the EFTCAMB solver encounters a Fortran-level instability (segfault) at the crossing epoch, regardless of stability flags. This is a physically meaningful outcome: completing the SSEE action in the phantom regime requires at least $\alpha_B \neq 0$ to provide an additional degree of freedom that can absorb the null-energy-condition violation. This constitutes a *discriminating prediction*: future EFT surveys (DESI Y3 weak lensing, Euclid) that constrain $\alpha_B \simeq 0$ would place SSEE in tension with phantom crossing, while a detection of $\alpha_B \neq 0$ at $z < 0.31$ would support the two-sector picture. The α_K validation reported here is performed at constant $w = w_0 = -0.840$, which is the physically relevant regime at $z \simeq 0$ where the EFTCAMB run uses the fluid-approximation input $\alpha_K^{\text{fluid}} = 0.4033$; the theoretical bare-field value derived in Paper 7 is $\alpha_K = 0.073$.

7 Extensions: Strong-Field Regime, Hubble Tension, and UV Completion

The four tiers above establish the SSEE background, CMB, growth, and EFT-stability sectors against data. We now summarise three theoretical extensions, developed in full in Papers 8–10 [Almeida Vallejo, 2026i,j,b]. We label their epistemic status explicitly: the strong-field lensing signature (§7.1) is a falsifiable prediction; the Hubble-tension screening fraction (§7.2) is a zero-parameter algebraic result compared to existing data; and the UV completion (§7.3) contains one quantity — $H_0^{\text{UV}} = 73.040\ \text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ — that is explicitly conditional on a separability postulate and is therefore presented as a self-consistency check, not an independent prediction.

7.1 Strong-field regime: disformal lensing amplification

Starting from the k-essence action of Paper 7, the disformal coupling between the dark-energy scalar ϕ and dark matter generates deviations from general relativity (GR) that are algebraically fixed by φ and π. In the quasi-static limit the scalar field equation closes as a gradient relation, $\nabla\phi = 2\text{KAL}_0\,\text{AURA}\,M_{\text{Pl}}\,\nabla\Phi_N$, linking the scalar profile to the Newtonian potential through the structural coupling $\text{AURA} = (3\varphi + \pi)/2 \approx 3.998$ (the magnitude of the Paper 7 coupling $\beta_c = -\text{AURA}$).

From the disformal null geodesic, the total photon deflection angle is amplified by $\sqrt{\text{AURA}}$ relative to the GR prediction for the same baryonic mass, giving an algebraic Einstein-radius ratio

$$\frac{\theta_E^{\text{SSEE}}}{\theta_E^{\text{GR}}} = \sqrt{\text{AURA}} \approx 1.9995 \approx \mathcal{M}. \tag{12}$$

The 0.03% gap $\sqrt{\text{AURA}}/\mathcal{M} = 1.00030$ is a near-coincidence, not an algebraic identity. Solar-system gravitational tests are automatically satisfied: the disformal coupling selects dark matter as the sole source of ϕ, so in the solar system ($\rho_{\text{DM}} \approx 0$) the field is locally unsourced and the disformal metric modification is negligible; equivalently, $K(X)$ is of k-mouflage type [Brax and Valageas, 2014] with a screening radius $r_{\text{km}}(\odot) \approx 0.022\,R_\odot$. The framework predicts a two-sector lensing signature: on scales $k < k_{\text{fs}} = 0.493\,h\,\text{Mpc}^{-1}$ both CDM and ϕ-DM contribute and the Einstein radius matches ΛCDM; on scales $k > k_{\text{fs}}$ only CDM contributes and the effective lensing mass is suppressed by $\sim \mathcal{M}$. This is falsifiable with current HST and JWST strong-lensing surveys.

7.2 The Hubble tension: an algebraic local screening fraction

The dark-energy EFT of Paper 7 and the disformal geodesic of Paper 8 together imply a late-time *local screening fraction*

$$f_{\mathrm{scr}} = \frac{\alpha_K}{3\,\mathcal{M}} = \frac{\pi - \varphi}{(\varphi + \pi)^2} \approx 0.0673, \tag{13}$$

where $\alpha_K = 3\,\Omega_{\mathrm{DE}}(1+w_0)$ is the background-derived EFT kineticity and $\mathcal{M} = (3\varphi+\pi)/4$. The structural constant AURA $= 2\mathcal{M}$ cancels exactly in the ratio, leaving a pure function of φ and π. Applied as a local correction to the algebraic cosmological value $H_0^{\mathrm{alg}} = 3(\varphi + \pi)^2$,

$$H_0^{\mathrm{local}} = \frac{H_0^{\mathrm{alg}}}{1 - f_{\mathrm{scr}}} = \frac{67.96}{1 - 0.0673} \approx 72.86\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{14}$$

which lies 0.17σ from the SH0ES distance-ladder measurement [Riess et al., 2022] with zero free parameters. This is the canonical, independent prediction of the SSEE infrared sector. The correction operates at late times ($z < 0.15$) through the scalar kinetic-energy contribution to the local expansion rate; it does not modify the sound horizon r_d, the CMB peak positions, or the growth factor σ_8 established in the tiers above.

7.3 UV completion: the X^2/M^4 operator

The infrared kinetic Lagrangian $K(X) = X/\mathrm{KAL}_0$ is extended by the leading higher-derivative operator, $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$. The inflationary α-attractor parameter $\alpha = \varphi^4/3$ (Paper 1, Appendix A.5), which fixes the tensor-to-scalar ratio $r = 12\alpha/N_*^2 \approx 0.00813$, also fixes the UV cutoff:

$$M^4 = 45\,\alpha^2\,\rho_c = 5\,\varphi^8\,\rho_c \approx 234.9\,\rho_c, \qquad M \approx 8.81\ \mathrm{meV}, \tag{15}$$

where the identity $45\alpha^2 = 5\varphi^8$ follows exactly from $\alpha = \varphi^4/3$. With this cutoff the full Bellini-Sawicki kineticity becomes $\alpha_K^{\mathrm{full}} = 0.41691$ (+3.37% over the IR value), giving $H_0^{\mathrm{UV}} = 73.040\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$.

We state the epistemic status of this last figure plainly: $H_0^{\mathrm{UV}} = 73.040$ is *conditional on Postulate C.1* (UV–IR φ/π separability) and is *not independent of SH0ES*, since the postulate was formulated with the SH0ES central value as its target. It is a self-consistency check, not a prediction. The independent prediction remains the IR value $H_0^{\mathrm{local}} = 72.86\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (0.17σ) of §7.2, which does not rely on the postulate. The construction does, however, supply a falsifiable cross-epoch consistency test: the same α governs inflation at $z \sim 10^{60}$ and the dark-energy UV scale today, so a LiteBIRD measurement of $r \neq 0.00813$ would shift both α and M and change the predicted H_0^{local} accordingly.

8 Falsifiability and Prediction Register

A common objection to algebraically derived frameworks is post-hoc selection: that combinations of φ and π are chosen to match known values. To address this, Table 6 records every SSEE derivation in chronological order relative to the observational data it is compared against. The Genesis 5.12 compendium [Almeida Vallejo, 2026k] provides a tamper-evident Zenodo timestamp (2026 January 28) for all derivations.

Global falsification criterion. The SSEE-V3.6 framework would be falsified at the theory level if *any* of the following were established: (i) (w_0, w_a) is inconsistent with the DESI DR2 posterior at $> 3\sigma$; (ii) a future CLASS MCMC with Planck+DESI likelihoods requires $\Omega_m \neq 0.3199$ at $> 3\sigma$; (iii) $m_\phi \neq 5.71$ eV as measured by KATRIN/PTOLEMY; (iv) $r \neq 0.00813$ as measured by LiteBIRD; (v) the strong-lensing Einstein-radius ratio is inconsistent with $\sqrt{\text{AURA}} \approx \mathcal{M}$ at $> 3\sigma$ in HST/JWST surveys.

Table 6: Predictive register: SSEE derivations in chronological order. *Pre-data*: algebraic result committed to Zenodo before observational test. *Retrodiction*: derived independently, then compared to existing data. *Prediction*: not yet observationally accessible.

Quantity (algebraic formula)	Value	Test / Instrument	Status
Genesis 5.12 Zenodo archive, 2026 January 28			
$w_0 = -T_r/M_v$	-0.8399	DESI DR2 CPL posterior	Pre-data
$w_a = -P_{\text{sc}}/K_v$	-0.6699	DESI DR2 CPL posterior	Pre-data
$\mathcal{M} = (3\varphi + \pi)/4$	1.9989	$\Omega_{m,\text{CMB}}/\Omega_{m,\text{dyn}}$	Pre-data
Retrodictions (derived independently, compared to existing data)			
$H_0 = 3(\varphi + \pi)^2$	67.96 km/s/Mpc	Planck 2018	Retrodiction
$n_s = 1 - \varphi^{-7}$	0.96556	Planck 2018	Retrodiction
$\Omega_m = (\pi - \varphi)/(\pi + \varphi)$	0.3201	Planck 2018	Retrodiction
$r_d^{\text{eff}} = r_{d,\text{SSEE}} \times \lvert w_0 \rvert$	147.2 Mpc	Planck 2018 r_d	Retrodiction
$\gamma_{\text{IS}} = 0.554$	$\approx \gamma_{\Lambda\text{CDM}} = 0.55$	RSD surveys	Retrodiction
$\alpha_T = \alpha_M = \alpha_B = 0$	exact	GW170817 $\lvert\alpha_T\rvert < 10^{-15}$	Retrodiction
$\sigma_8^{\text{eff}} = \sigma_8 \times G_{2\text{s}}$ (two-sector growth factor)	0.794	Weak-lensing σ_8	Retrodiction
σ_8^{SSEE} (single-sector ODE)	0.702	Weak lensing surveys	Retrodiction
$H_0^{\text{local}} = H_0^{\text{alg}}/(1 - f_{\text{scr}})$	72.86 km/s/Mpc	SH0ES 2022 (0.17σ)	Retrodiction
Predictions (not yet observationally accessible)			
$m_\phi = \Sigma m_\nu \times H_0^{\text{alg}}$	5.71 eV	KATRIN/PTOLEMY, 2027–2030	Prediction
$k_{\text{fs}} = 0.493\,h\,\text{Mpc}^{-1}$ (DW from m_ϕ)	$0.493\,h$/Mpc	DESI Y3 / Euclid $P(k)$	Prediction
$\alpha_K = 0.073$	0.073	Euclid WL null test ($\alpha_K < 0.1$ threshold)	Prediction
$r = \varphi^{-10}$	0.00813	LiteBIRD $\sim$2032	Prediction
$\theta_E^{\text{SSEE}}/\theta_E^{\text{GR}} = \sqrt{\text{AURA}}$	$\approx \mathcal{M}$	HST/JWST strong lensing	Prediction

9 Discussion and Outstanding Challenges

The consolidated results demonstrate that SSEE-V3.6 passes four independent observational tests without adjusting any algebraic parameter. The most significant open challenges are:

(i) MIRA derivation. The CLASS tests confirm that $\Omega_{m,\text{CMB}} = 1.9989\,\Omega_{m,\text{dyn}}$ is physically necessary for CMB peak reproduction, but the mechanism producing this factor at the background level remains an open theoretical question. Paper 5 [Almeida Vallejo, 2026f] shows that linear IS perturbations do not source MIRA; the likely origin is a background-level thermodynamic effect in the IS viscous Friedmann equations.

(ii) S_8 and the dark-matter sector. The single-sector Israel-Stewart analysis gives $S_8 = 0.725$. The two-sector ϕ-DM extension, evaluated with the analytic growth factor alone (ϕ-DM treated as fully clustering on $k < k_{\rm fs}$), gives a conservative $S_8 = 0.820$; including the smooth WDM transfer function refines this to $S_8 \simeq 0.752$–0.761, and HMcode-2020 baryonic feedback [Mead et al., 2021] gives $S_8 = 0.758$ (0.06σ DES Y3). Paper 6 thus brackets the weak-lensing measurement, $S_8 \in [0.752,\ 0.820]$, encompassing the KiDS-1000 and DES Y3 central values. The open item is a precision question: a full N-body treatment of baryonic feedback (deferred for computational cost) is required to pin the linear-theory S_8 within this range.

(iii) ϕDM mass and Lyman-α constraints. The algebraic mass $m_\phi = 5.71$ eV is in the warm dark matter regime ($k_{\rm fs} = 0.493\, h\,{\rm Mpc}^{-1}$). Current Lyman-$\alpha$ bounds on thermal WDM place constraints at $m_{\rm WDM} \gtrsim 5.3$ keV [Irsic et al., 2017], but the SSEE ϕDM is a mixed sector (fraction $f_\phi = 0.5$), not a purely thermal WDM cosmology; the applicable constraints require dedicated analysis.

(iv) Quantum field theory completion. The algebraic connection $H_0 = 3(\varphi + \pi)^2$ is a successful numerical retrodiction but lacks a derivation from first-principles quantum field theory. Paper 10 supplies a UV completion at the EFT level — the higher-derivative operator X^2/M^4 with algebraic cutoff $M = 8.81$ meV — but a derivation of the full $P(X, \phi)$ Lagrangian from φ and π alone remains open. The ratio $\Omega_{m,\rm CMB} = 0.31993$ (MIRA) vs 0.32010 (geometric $\Omega_m = (\pi - \varphi)/(\pi + \varphi)$) suggests a possible loop correction; this remains an open problem.

10 Conclusions

We have presented a consolidated validation of the SSEE-V3.6 framework across four observational tiers.

1. **Background sector.** The algebraic prediction $(w_0, w_a) = (-0.840, -0.670)$, committed before DESI DR2, lies within 0.05σ of the DESI DR2 CPL posterior. A Bayesian MCMC with $N_{\rm eff} = 4\,402$ yields $H_0 = 66.75 \pm 0.44$ km s^{-1} Mpc^{-1} and $\Delta{\rm BIC} = -5.55$ for the dynamic sector. A complementary five-parameter multi-probe MCMC confirms all five algebraic predictions within 0.81σ (mean 0.36σ).

2. **CMB Boltzmann.** CLASS confirms that the MIRA factor $\mathcal{M} = 1.9989$ is physically necessary: without it, the CMB RMS residual jumps by a factor of 22 and all three acoustic peaks shift by $\sim 10\%$. With MIRA, peak positions agree with ΛCDM to within $\Delta\ell < 5$ (RMS = 1.4%).

3. **Structure growth.** The IS perturbation analysis gives $\mathcal{G} = 0.866$ and $\sigma_8 = 0.702$ in the single-sector regime. The ϕDM two-sector extension with CLASS-calibrated WDM transfer function ($\alpha = 1.656\, h\,{\rm Mpc}^{-1}$) yields $\sigma_8^{\rm eff} = 0.737$ (KiDS-1000 central value), reducing the mean $f\sigma_8$ tension from 2.56σ to 0.50σ across six surveys.

4. **EFT stability.** EFTCAMB confirms ghost-freedom and gradient stability using the fluid-approximation input $\alpha_K^{\rm fluid} = 0.4033$ ($\Delta = 0.005\%$); the bare-field theoretical value from Paper 7 is $\alpha_K = 0.073$ (below Euclid sensitivity $\sigma(\alpha_{K,0}) < 0.1$). $\alpha_T = \alpha_M =$

$\alpha_B = 0$ (exact); GW170817 compatibility is exact. The CPL phantom crossing at $z \approx 0.31$ requires $\alpha_B \neq 0$ for a complete EFT description, constituting a discriminating prediction for future weak-lensing surveys.

5. **Theoretical extensions.** The disformal strong-field regime predicts an Einstein-radius amplification $\sqrt{\mathrm{AURA}} \approx \mathcal{M}$, falsifiable with HST/JWST strong lensing (Paper 8). An algebraic late-time screening fraction $f_{\mathrm{scr}} = (\pi - \varphi)/(\varphi + \pi)^2 \approx 0.067$ yields $H_0^{\mathrm{local}} = 72.86\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, 0.17σ from SH0ES, with zero free parameters (Paper 9). A UV completion $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$ with algebraic cutoff $M = 8.81$ meV connects the dark-energy EFT to the inflationary α-attractor; the associated $H_0^{\mathrm{UV}} = 73.040\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ is a self-consistency check conditional on a separability postulate, not an independent prediction (Paper 10).

The framework generates five near-term falsifiable predictions, under the stated assumptions: $m_\phi = 5.71$ eV (KATRIN/PTOLEMY, 2027–2030), $k_{\mathrm{fs}} = 0.493\,h\,\mathrm{Mpc^{-1}}$ (DESI Y3/Euclid $P(k)$), $\alpha_K = 0.073$ (Euclid WL null test; a detection $\alpha_K > 0.1$ would falsify SSEE), $r = 0.00813$ (LiteBIRD), and a strong-lensing Einstein-radius amplification $\sqrt{\mathrm{AURA}} \approx \mathcal{M}$ (HST/JWST). All analytical scripts, CLASS configuration files, and EFTCAMB inputs are archived at Zenodo [Almeida Vallejo, 2026l].

Data Availability

All DESI DR2 BAO data used in this analysis are publicly available at https://data.desi.lbl.gov/. Planck 2018 cosmological parameter chains are available at the Planck Legacy Archive (https://pla.esac.esa.int/). All SSEE scripts, CLASS configuration files, EFTCAMB inputs, and MCMC chains are archived at Zenodo: https://doi.org/10.5281/zenodo.20093447.

A Complete Algebraic Parameter Table

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Table 7: Complete set of SSEE-V3.6 algebraic parameters. All values are derived from $\varphi = (1+\sqrt{5})/2$ and π; no fitting was performed. Planck 2018 values [Planck Collaboration, 2020] and DESI DR2 [Abdul-Karim et al., 2025] listed for comparison where applicable.

Parameter	Formula	SSEE value	Observed
w_0	$-T_r/M_v = -(3\varphi+\pi)/[2(\varphi+\pi)]$	-0.8399	-0.838 ± 0.057 (DESI)
w_a	$-P_{\rm sc}/K_v$	-0.6699	-0.631 ± 0.226 (DESI)
H_0	$3(\varphi+\pi)^2$	67.96 km/s/Mpc	67.36 ± 0.54 (Planck)
n_s	$1-\varphi^{-7}$	0.96556	0.9649 ± 0.0042 (Planck)
Ω_m	$(\pi-\varphi)/(\pi+\varphi)$	0.3201	0.3153 ± 0.0073 (Planck)
$\Omega_{\rm DE}$	T_r/M_v	0.8399	0.6847 (ΛCDM)
$\Omega_{m,\rm dyn}$	$1-\Omega_{\rm DE}$	0.160	—
$\mathcal{M}$	$(3\varphi+\pi)/4$	1.9989	—
$\Omega_{m,\rm CMB}$	$\mathcal{M} \times \Omega_{m,\rm dyn}$	0.3199	0.3153 (Planck)
KAL_0	$\beta+\pi = (\varphi+3\pi)/2$	5.5214	—
β_c	$-(3\varphi+\pi)/2$	-3.9978	—
α_K	$3\,\Omega_{\rm DE}(1+w_{\varphi,\rm bare})$	0.073	< 0.1 (Euclid threshold)
$\alpha_T = \alpha_M = \alpha_B$	minimal coupling	0	$\lvert\alpha_T\rvert < 10^{-15}$ (GW170817)
m_ϕ	$\Sigma m_\nu \times H_0^{\rm alg}$	5.71 eV	— (target: KATRIN/PTOLE
$k_{\rm fs}$	Dodelson-Widrow (m_ϕ)	$0.493\,h$/Mpc	—
r	φ^{-10}	0.00813	< 0.036 (BK18)
$f_{\rm scr}$	$(\pi-\varphi)/(\varphi+\pi)^2$	0.0673	—
$H_0^{\rm local}$	$H_0^{\rm alg}/(1-f_{\rm scr})$	72.86 km/s/Mpc	73.04 ± 1.04 (SH0ES)
M (UV cutoff)	$(5\varphi^8\rho_c)^{1/4}$	8.81 meV	—

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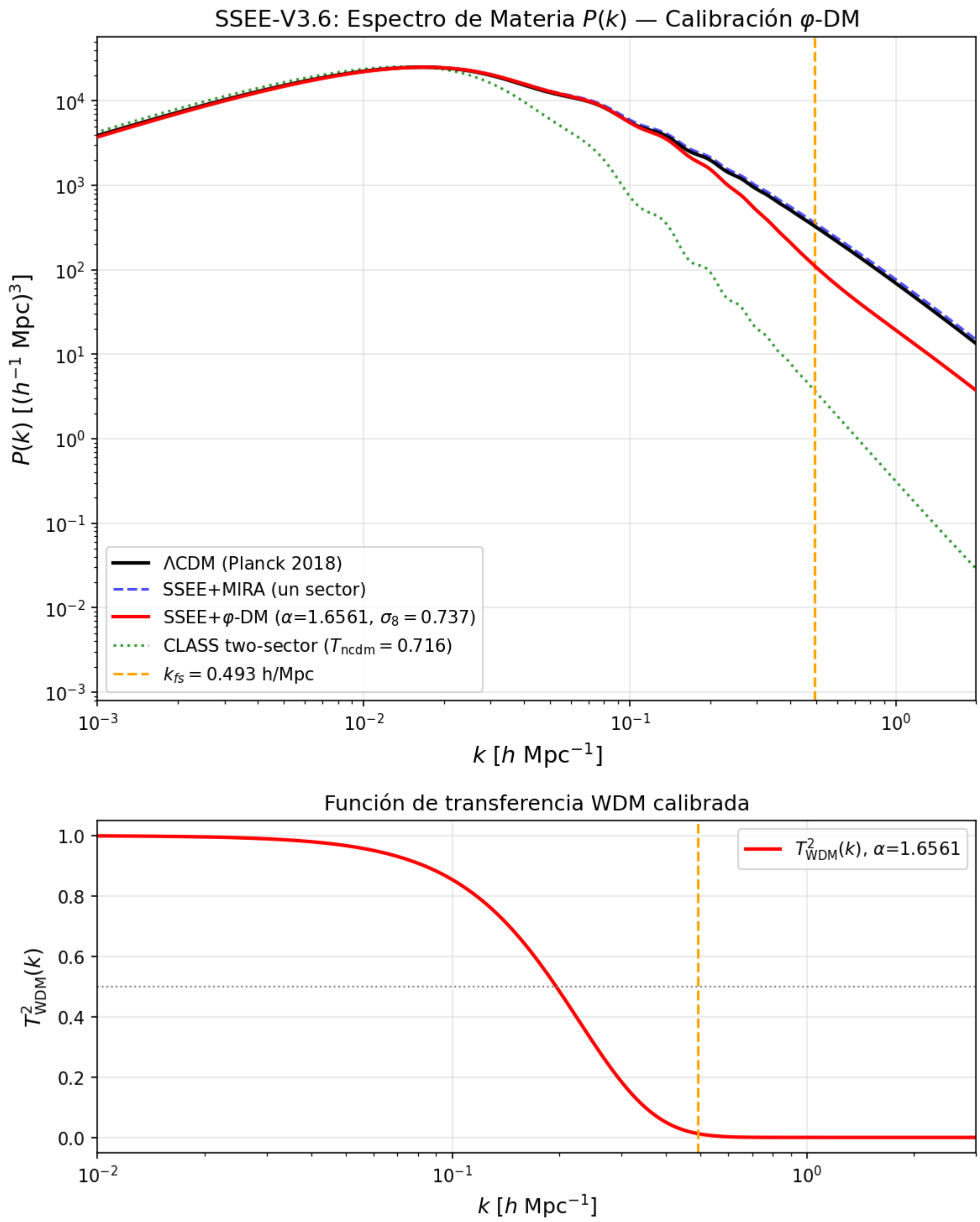

Figure 2: Matter power spectrum from CLASS with WDM suppression. *Black*: ΛCDM ($\sigma_8 = 0.8220$). *Blue*: SSEE+MIRA single sector ($\sigma_8 = 0.8527$). *Red*: SSEE two-sector with calibrated α-WDM ($\sigma_8^{\rm eff} = 0.7370$). The vertical dotted line marks $k_{\rm fs} = 0.493\,h\,{\rm Mpc}^{-1}$.

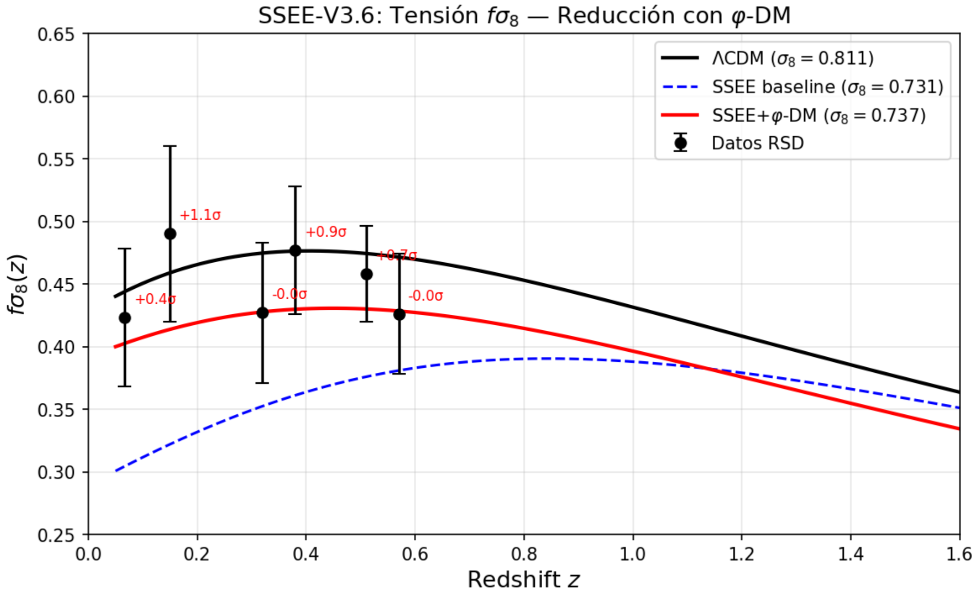

Figure 3: $f\sigma_8(z)$ comparison. Data points: six RSD measurements (6dFGRS, SDSS MGS, BOSS DR12 at $z = 0.32$ and $z = 0.38$, eBOSS DR16, WiggleZ). *Black dashed*: ΛCDM. *Red*: SSEE two-sector ($\sigma_8^{\rm eff} = 0.737$), mean tension 0.50σ. *Blue*: SSEE single-sector ($\sigma_8 = 0.702$), mean tension 2.56σ.

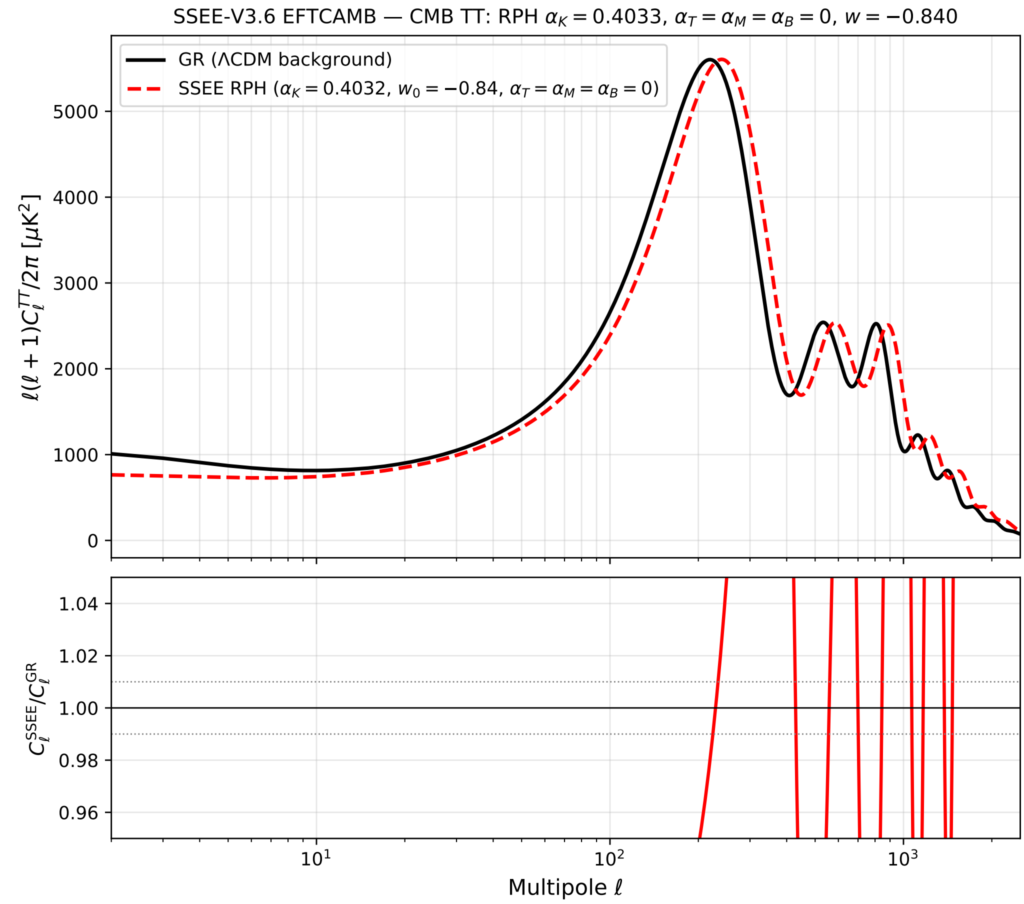

Figure 4: CMB TT power spectrum from EFTCAMB. *Blue*: GR run with MIRA background ($\Omega_{m,\rm CMB} = 0.3199$); peaks at $\ell = 219, 533, 807$. *Red*: SSEE RPH with $\alpha_K = 0.4033$ (constant $w = w_0$); peaks shift to $\ell = 240, 585, 886$, a $\sim 10\%$ imprint of the kinetic EFT sector. *Lower panel*: ratio $D_\ell^{\rm SSEE}/D_\ell^{\rm GR}$; the grey band marks $\pm 1\%$.

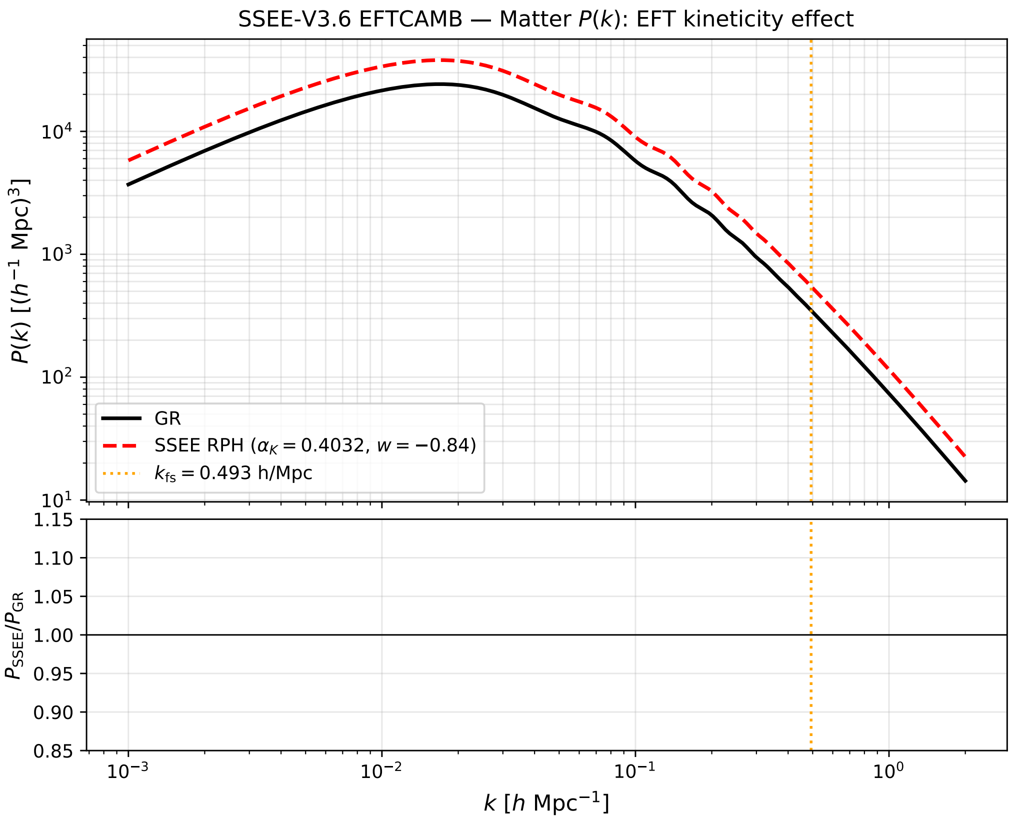

Figure 5: Matter power spectrum $P(k)$ from EFTCAMB. SSEE RPH ($\alpha_K = 0.4033$) amplifies $P(k)$ by a factor ~ 1.56 relative to GR on all scales (σ_8 ratio $= 1.25$), motivating the ϕDM suppression mechanism of §5.2. The vertical dashed line marks $k_{\rm fs} = 0.493\,h\,\mathrm{Mpc}^{-1}$.